

Remanufacturing scheduling with collaborative-learning modes and human factors

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Why this problem?

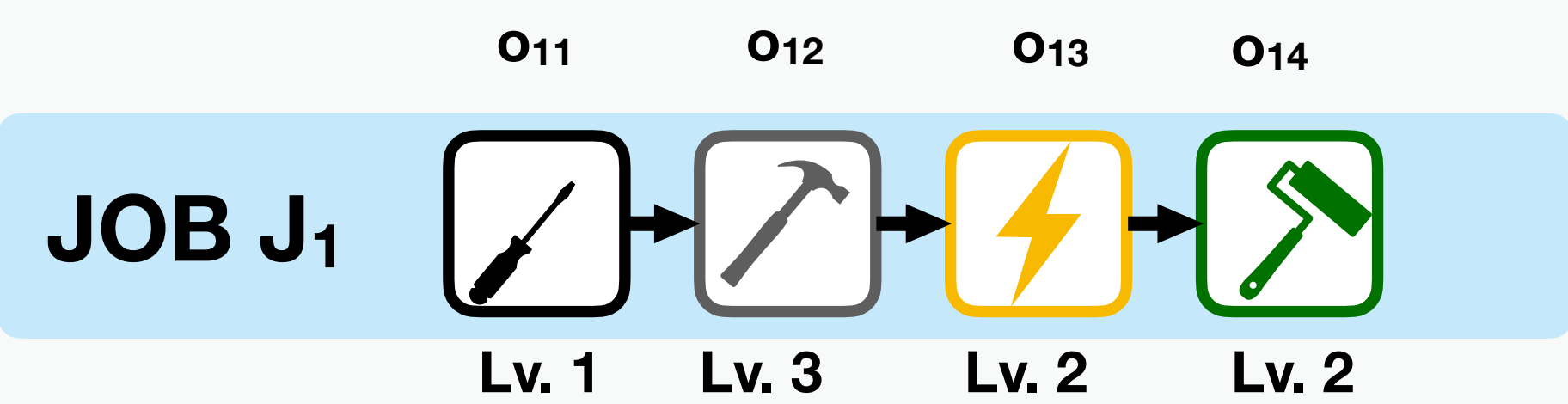
Remanufacturing keeps products in use—but its uncertain operations still depend on skilled people.

- Workers have heterogeneous and evolving skills
- Training can occur during production
- Collaboration changes time and cognitive effort

Hypothesis :

- The cognitive workload associated with solo mode is not considered

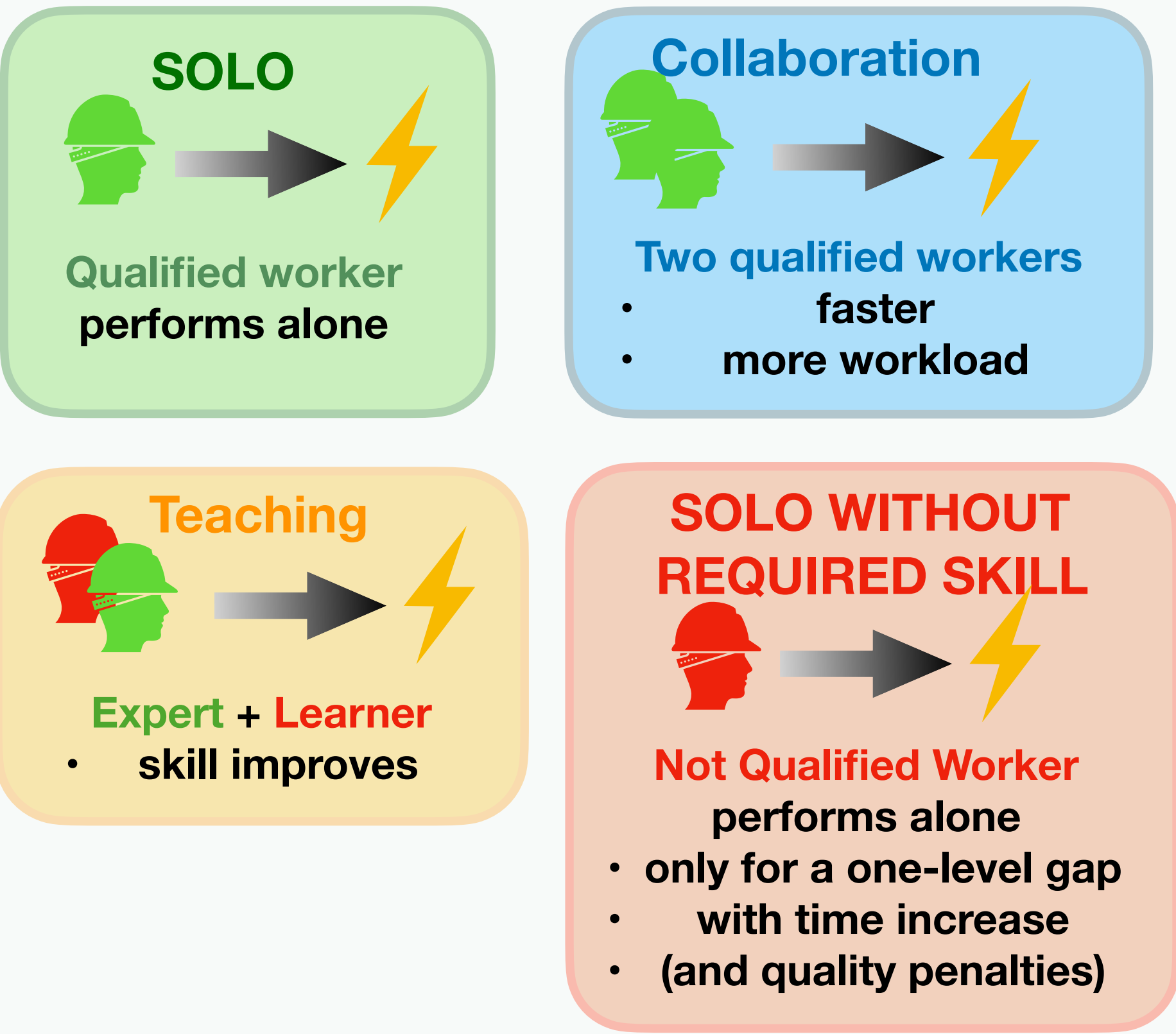
What must be scheduled?



Operation O₁₃ of J₁
Electrial - skill level 2

Worker	Electrical skill	Execution Mode
 W ₁	Level 1	Teaching - Alone No level
 W ₂	Level 2	Alone
 W ₃	Level 1	Teaching - Alone No level

Execution modes



Lexicographic objectives

1 $\sum_{i \in J} (w_i \cdot job_done_i + \sum_{j \in j_i} task_done_{ij})$ 

2 $\sum_{k \in W} \sum_{m \in M} skills_{km}$ 

3 $\sum_{k \in W} \sum_{m \in M} CL_{km}$ 

Exact method: MILP

INPUTS
Jobs · operations · worker skills · processing times · difficulty

MULTI-OBJECTIVE MILP
Decisions
assignment · execution mode · operation sequence
Constraints
precedence · no overlap · skills & learning · cognitive workload

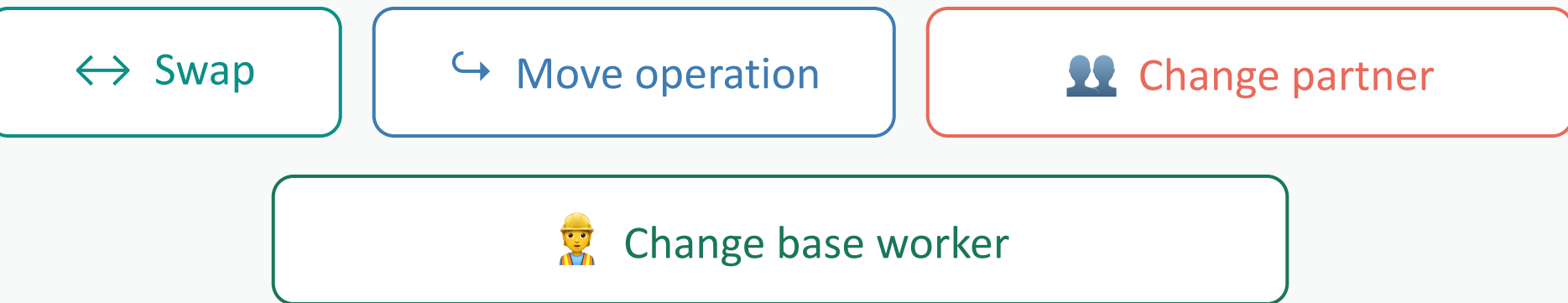
GUROBI - Lexicographic optimization

Scalable method - Metaheuristic

A compact solution encoding links operation order and worker assignments.

	O ₂₁	O ₁₁	O ₂₂	O ₁₂	O ₂₃	O ₁₃	
OSC	2	1	1	2	1	1	← job indices
FWAC	1	1	1	1	2	2	← First worker indices
SWAC	-1	-1	3	2	-1	-1	← Second worker indices
	A	A	C	T	A	T	← Execution Mode
BWAC	1				2		← Base worker indices
	J ₁				J ₂		

Neighborhood operators



ONE MODEL · TWO SOLUTION SCALES

MILP validates optimality on small instances; the metaheuristic targets larger instances.

Results

Lorem ipsum

① Loremp ipsum

Lorem ipsum

INSERT FINAL FIGURE

② Lorem ipsum

Key takeaways